
ESG Disclosure Quality of Pension Funds in East Asia: Evidence from China, South Korea and Japan

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Abstract: This study develops a Pension Fund Disclosure Quality Index (PFDQI) to evaluate the ESG disclosure quality of three major East Asian public pension funds—China’s National Council for Social Security Fund (NCSSF), South Korea’s National Pension Service (NPS), and Japan’s Government Pension Investment Fund (GPIF)—over the period 2020–2025. By using content analysis and sigma-convergence analysis, the study examines both cross-national differences and dynamic changes in disclosure quality. The results indicate that ESG disclosure quality improved consistently across all three pension funds during the study period. GPIF maintained the highest disclosure quality, NPS demonstrated steady progress, and NCSSF exhibited a significant catch-up effect. It is also reflected that climate disclosure and credibility emerged as the weakest dimensions across the three institutions. The findings further reveal a gradual convergence trend in ESG disclosure quality, although substantial institutional differences remain. As a result, this study contributes to the sustainable finance literature by providing a pension fund-specific ESG disclosure assessment framework and new comparative evidence from East Asia.

Keywords: ESG disclosure quality; pension funds; responsible investment; sustainable finance; East Asia

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Received 15 April 2026, Revised 20 May 2026, Accepted 10 June 2026

1 Introduction

Environmental, Social, and Governance (ESG) considerations have emerged as a fundamental component of sustainable finance and long-term value creation in global capital markets. As climate-related risks, demographic aging, and sustainable development challenges intensify, pension funds—among the largest and most influential long-term institutional investors worldwide—are facing growing expectations to contribute actively to sustainable finance and responsible investment. According to the Organization for Economic Co-operation and Development (OECD), global pension assets have reached unprecedented levels and are undergoing a gradual transition toward ESG-oriented investment strategies. Given their long investment horizons, extensive beneficiary coverage, and fiduciary responsibilities, pension funds play a critical role not only in capital allocation and financial market stability but also in supporting the transition toward a low-carbon and sustainable economy.

East Asia represents one of the most dynamic regions in terms of pension asset growth and sustainable finance development. The Government Pension Investment Fund of Japan (GPIF), the National Pension Service of South Korea (NPS), and the National Council for Social Security Fund of China (NCSSF) are among the region's most influential public pension institutions, which embedded within distinct governance structures and institutional environments. Although all three institutions have integrated ESG considerations into their investment frameworks, substantial differences remain exist. As ESG regulations, sustainable finance initiatives, and disclosure standards continue to evolve across East Asia, important questions arise regarding the trajectory of ESG disclosure among public pension funds. In particular, it remains unclear whether disclosure quality is converging toward internationally recognized standards or whether institutional differences continue to generate divergent outcomes across national contexts. Addressing these questions is crucial not only for understanding the role of pension funds in advancing sustainable finance but also for explaining the broader institutional dynamics shaping ESG governance in the region.

Despite the growing importance of ESG disclosure within the pension fund sector, existing scholarship has focused predominantly on corporate sustainability reporting, while systematic assessments of ESG disclosure quality among pension funds remain limited. Moreover, current ESG evaluation frameworks are largely designed for corporate entities and may not adequately capture the distinctive characteristics of pension funds as long-term asset owners and fiduciary institutions. Consequently, there remains a lack of comprehensive and comparable measures for evaluating ESG disclosure quality within the public pension sector, particularly in cross-national settings.

To address these limitations, this study examines three major public pension funds in East Asia—the National Council for Social Security Fund of China (NCSSF), the National Pension Service of South Korea (NPS), and the Government Pension Investment Fund of Japan (GPIF). By drawing on a newly developed Pension Fund Disclosure Quality Index (PFDQI), the study systematically evaluates the quality of ESG-related disclosure across these institutions over the period 2020–2025 and investigates patterns of cross-national variation as well as their dynamic evolution over time. The remainder of this article is organized as follows. Section 2 reviews the relevant literature and develops the analytical framework. Section 3 outlines the research design and presents the construction of the Pension Fund Disclosure Quality Index (PFDQI). Section 4 reports and discusses the research findings. Finally, Section 5 concludes by summarizing the main findings, highlighting policy implications, and identifying limitations for future research.

2 Literature Review

2.1 ESG Disclosure Research

Traditionally, the literature on ESG disclosure was primarily informed by information asymmetry theory, which emphasizes the role of disclosure in mitigating informational imbalances between managers and external stakeholders. Dhaliwal et al. (2011) find that firms voluntarily issuing sustainability reports generally experience lower equity financing costs, suggesting that ESG disclosure can generate tangible economic benefits. Similarly, Cormier et al. (2011) report that higher-quality environmental disclosure contributes positively to market valuation by improving investor perceptions of

corporate transparency. Cheng et al. (2014) further demonstrate that enhanced ESG transparency can alleviate financing constraints by reducing uncertainty and strengthening stakeholder confidence.

As ESG investing has expanded rapidly over the past decade, researchers have increasingly emphasized disclosure quality as a more informative measure. Michelon et al. (2015) argue that disclosure quantity should not be conflated with disclosure quality, emphasizing that meaningful disclosure requires materiality, comparability, completeness, and verifiability. Berg et al. (2022) reveal substantial divergence among major ESG rating agencies, including MSCI, Sustainalytics, and Refinitiv, highlighting persistent challenges in ESG measurement and evaluation. The emergence of international reporting standards has further accelerated the institutionalization of ESG disclosure. Frameworks such as the GRI, SASB, TCFD (2017), and ISSB (IFRS Foundation, 2023) have progressively shifted sustainability reporting toward greater standardization and comparability. Nevertheless, the applicability of these frameworks to institutional investors remains insufficiently explored, particularly in the context of public pension funds.

2.2 Pension Funds and Responsible Investment

Pension funds constitute one of the most influential categories of long-term institutional investors in global capital markets. Richardson (2009) argues that pension funds possess unique potential to influence long-term economic and environmental outcomes because of their intergenerational investment horizons and fiduciary responsibilities. Following the widespread adoption of the Principles for Responsible Investment (PRI), responsible investment has increasingly become embedded within pension fund governance structures and investment processes (PRI, 2023).

Clark & Hebb (2014) analyzed that pension funds are widely recognized as key actors in advancing sustainable finance and promoting responsible investment practices. The literature on pension fund responsible investment has primarily focused on three areas. First, Friede et al., (2015) examined the financial implications of ESG integration, suggested that the incorporation of ESG factors can enhance portfolio resilience, improve long-term risk-adjusted performance, and mitigate exposure to sustainability-related risks. Second, Clark et al., (2017) explored the role of governance structures, stated that pension funds with stronger governance arrangements, greater transparency, and clearer fiduciary mandates are more likely to implement systematic ESG investment strategies. Third, Dimson et al., (2015) highlighted the growing importance of active ownership, emphasized that pension funds can influence corporate ESG performance through shareholder voting, engagement activities, and stewardship initiatives. In addition, stewardship has emerged as a central pillar of responsible investment. Dimson et al. (2020) further demonstrate that sustained shareholder engagement can produce measurable improvements in corporate ESG performance, reinforcing the importance of stewardship activities as a mechanism through which pension funds influence sustainability outcomes.

2.3 ESG Disclosure Quality and Pension Funds

More recently, research has expanded investment behavior to consider how pension funds communicate their ESG-related activities and commitments. Hebb (2018) demonstrated that effective disclosure is increasingly viewed as an essential component of responsible investment because it enhances transparency, strengthens accountability, and enables stakeholders to assess whether pension funds are fulfilling their fiduciary and sustainability obligations. Other research has expanded investment behavior to consider how pension funds communicate their ESG-related activities and commitments. Liang & Renneboog (2017) stated that pension funds with more developed ESG governance structures, stronger stewardship frameworks, and greater institutional commitment to sustainability generally provide disclosures that are more comprehensive, systematic, and verifiable. Nakai (2020) & Sato (2022) constructed the research on Japan's GPIF, which highlights its leadership in responsible investment reporting, stewardship transparency, and climate-risk disclosure. Similarly, Kim & Lee (2021) & Park et al., (2023) studied on South Korea's NPS document and revealed the rapid expansion of ESG integration and responsible investment disclosure practices. In China, scholars have increasingly examined ESG governance and sustainability-related disclosure among public pension institutions, Wang et al., (2022) & Liu & Zhang (2023) emphasis the evolving role of the National Council for Social Security Fund in promoting responsible investment.

Despite the growing literature on responsible investment, limited attention has been paid to the systematic evaluation of ESG disclosure quality among pension funds, particularly in cross-national contexts. To address these limitations, this study develops a Pension Fund Disclosure Quality Index (PFDQI) that evaluates ESG disclosure quality across five dimensions—transparency, completeness, climate disclosure, responsible investment, and credibility—and applies this framework to compare the ESG disclosure practices of the NCSSF, NPS, and GPIF during the period 2020–2025.

3 Research method

3.1 Research Design

This study employs a comparative case study design combined with content analysis to examine the ESG disclosure quality of three major public pension funds in East Asia: the National Council for Social Security Fund of China (NCSSF), the National Pension Service of South Korea (NPS), and the Government Pension Investment Fund of Japan (GPIF). Covering the period from 2020 to 2025, the study seeks not only to assess the evolution of ESG disclosure quality but also to identify the institutional conditions that contribute to variations in disclosure practices. Comparative case studies are particularly valuable for exploring how organizational outcomes are shaped by broader governance arrangements, regulatory structures, and socio-economic contexts while maintaining analytical control over case selection (Yin, 2018). Content analysis, meanwhile, provides a systematic, transparent, and replicable method for extracting, coding, and quantifying information from publicly available documents (Krippendorff, 2018). The integration of these methods has been widely adopted in sustainability reporting and ESG research, offering a robust framework for evaluating disclosure quality across organizations operating in different institutional settings (Hess, 2007; Christofi et al., 2012). Case selection follows the logic of the Most Different Systems Design (MDSD) proposed by Przeworski and Teune (1970). Such variation provides a valuable opportunity to examine whether and how differences in institutional environments are associated with divergent ESG disclosure outcomes.

3.2 Construction of the Pension Fund Disclosure Quality Index (PFDQI)

To evaluate ESG disclosure quality among public pension funds, this study develops a dedicated assessment framework, the Pension Fund Disclosure Quality Index (PFDQI). The construction of the PFDQI is informed by four internationally recognized ESG disclosure frameworks: the Global Reporting Initiative (GRI) Standards, the Task Force on Climate-related Financial Disclosures (TCFD) recommendations, the International Sustainability Standards Board (ISSB) standards, and the Principles for Responsible Investment (PRI) reporting framework. These frameworks collectively provide the conceptual foundation for identifying the core dimensions of high-quality ESG disclosure and ensure consistency with prevailing international disclosure norms (GRI, 2021; TCFD, 2021; PRI, 2022; ISSB, 2023). Drawing upon these standards as well as the emerging literature on pension fund ESG disclosure and responsible investment governance (Garz & Volk, 2021; UNEP FI, 2020), the PFDQI comprises five first-order dimensions and eighteen second-order indicators, with a maximum score of 100 points. Table 1 presents the complete indicator structure and scoring protocol of the PFDQI.

Table 1. Pension Fund Disclosure Quality Index (PFDQI)

First-Order Dimension	Weight	Second-Order Indicators	Number of Indicators	Scoring Criteria
Transparency	20%	ESG report publication; Responsible investment report; ESG strategy statement; Quantitative target disclosure	4	Binary scoring: Full disclosure = 1; Partial disclosure = 0.5; No disclosure = 0
Completeness	25%	Coverage of Environmental (E) issues; Coverage of Social (S) issues; Coverage of Governance (G) issues; Quantitative ESG performance indicators	4	Binary scoring with weighted aggregation; additional scores assigned according to the depth of quantitative disclosure
Climate Disclosure	25%	Adoption of the TCFD framework; Climate-risk materiality assessment; Quantified carbon-reduction targets; Multi-scenario climate scenario analysis	4	Graduated scoring: Full disclosure = 1; Partial disclosure = 0.5; No disclosure = 0; scenario analysis weighted according to the number and sophistication of scenarios disclosed

Responsible Investment (RI)	20%	PRI signatory status and annual reporting; Stewardship and engagement activities; Public disclosure of shareholder voting records; Corporate engagement disclosure	4	Binary scoring; adjustments made according to the completeness of voting and engagement disclosures
Credibility	10%	Independent third-party assurance; Level of external assurance (limited or reasonable assurance)	2	Graduated scoring: Reasonable assurance = 1; Limited assurance = 0.5; No assurance = 0
Total	100%	18 Second-Order Indicators	18	Weighted composite score with a maximum of 100 points

***Note:** The weighting structure is determined based on both the relative importance of disclosure dimensions within internationally recognized ESG reporting frameworks and the distinctive governance characteristics of public pension funds. All second-order indicators are evaluated using standardized coding and scoring procedures to ensure cross-institutional and longitudinal comparability. The maximum attainable PFDQI score is 100 points.*

3.3 PFDQI scoring system and coding protocol

This study adopts a structured content analysis approach to operate and evaluate the indicators of the Pension Fund Disclosure Quality Index (PFDQI). To ensure methodological rigor, transparency, and replicability, a standardized coding protocol is implemented throughout the assessment process. The coding procedure consists of four stages,

The first stage is document collection and evidence identification. Based on the operational definitions of the PFDQI indicators, relevant qualitative statements, quantitative disclosures, and policy commitments are identified and recorded. Stage two is indicator coding. Followed by evidence identification, each indicator is independently evaluated using a standardized coding manual. As a result, scores are assigned according to predefined scoring criteria, including binary and graduated scoring rules specified in the PFDQI framework. After the initial coding process, stage 3 turns to reliability review, any discrepancies identified during the review process are resolved through reference to the coding protocol and operational definitions. Finally, stage 4 focuses on composite index construction. Indicator scores are aggregated using the predetermined weighting structure of the PFDQI. Table 2 presents the classification standards for selected core dimensions, including transparency, completeness, and climate disclosure, thereby providing a clearer basis for interpreting variations in disclosure quality across institutions and over time.

Table 2. Illustrative Tier Classification Standards for Core PFDQI Dimensions

Disclosure Tier	Transparency Score	Completeness Score	Climate Disclosure Score	Classification Criteria
High	≥75	≥75	≥75	Near-comprehensive indicator coverage; sufficient quantitative disclosures; consistent with international standards
Medium	50–74	50–74	50–74	Core indicators covered; quantitative disclosure limited or partially absent; disclosure practices at an improvement stage
Low	<50	<50	<50	Incomplete indicator coverage; predominantly qualitative in nature; absence of a systematic disclosure framework

Note: Tier thresholds are calibrated against a maximum possible score of 100 points for each dimension. Actual scores are computed by aggregating indicator-level values assigned through binary or graduated scoring rules; tier labels are not used as scoring inputs. The classification standards presented in this table serve as an interpretive aid to illustrate the scoring logic and do not constitute the basis for final score assignment.

3.4 σ -convergence

To examine whether ESG disclosure quality among East Asian pension funds exhibits a tendency toward convergence over time, this study employs sigma-convergence (σ -convergence) analysis to assess changes in the cross-sectional dispersion of Pension Fund Disclosure Quality Index (PFDQI) scores during the period 2020–2025. Following the conventional approach in the convergence literature, sigma-convergence is measured using the cross-sectional standard deviation of annual PFDQI scores. A downward trend in the standard deviation over time is interpreted as evidence of convergence, whereas an upward trend indicates divergence. The cross-sectional standard deviation is calculated as follows:

$$\sigma_t = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_{it} - \bar{X}_t)^2}$$

Where X_{it} denotes the PFDQI score of institution i in year t , $\bar{X}_t$ represents the cross-sectional mean score across the three pension funds in that year, and $N=3$ is the number of institutions in the sample. To strengthen cross-temporal comparability and control for the influence of mean-level fluctuations on dispersion estimates, this study additionally introduces the Coefficient of Variation (CV) as a supplementary measure, defined as:

$$CV_t = \frac{\sigma_t}{\bar{X}_t}$$

This metric provides a more stable characterization of relative dispersion across the sample under conditions of shifting mean levels, thereby enhancing the interpretive robustness of convergence trend assessments.

3.5 Data Source and Sample Description

All data utilized in this study are derived from publicly available documents released by three pension institutions under investigation, covering the period from 2020 to 2025. To enhance data reliability and source transparency, a hierarchical data collection protocol was adopted. Institutional annual reports, ESG reports, responsible investment reports, and other official disclosure documents constitute the primary data sources. PRI Transparency Reports and TCFD-related disclosure materials serve as secondary sources for cross-validation and supplementary evidence. Regulatory documents and policy reports issued by relevant governmental and supervisory authorities are employed as contextual references to facilitate institutional interpretation but are not directly incorporated into the PFDQI scoring process.

Table 3 summarizes the principal and supplementary data sources used for each pension institution. The use of multiple documentary sources reduces the risk of relying on a single disclosure channel, thereby enhancing the robustness and credibility of the assessment process.

Table 3. Primary Data Sources for the Three East Asian Pension Funds

Institution	Country	Primary Sources	Supplementary Sources
GPIF	Japan	GPIF Annual Reports (2020–2025); GPIF ESG Activity Reports; PRI Transparency Reports	Japan FSA Stewardship Code compliance reports; TCFD disclosure materials
NPS	South Korea	NPS Annual Reports (2020–2025); NPS Responsible Investment Reports; PRI Transparency Reports	Korea FSC Stewardship Code disclosures; NPS ESG webpages
NCSSF	China	National Social Security Fund Annual Reports (2020–2025); official NCSSF announcements and disclosures	People's Bank of China Green Finance Reports; CSRC ESG-related policy documents; PRI Transparency Reports

The study period of 2020–2025 was selected for three reasons. First, the year 2020 represents an important turning point in the global institutionalization of ESG disclosure. Second, a six-year observation window offers sufficient temporal variation to examine both longitudinal changes in ESG disclosure quality and potential convergence trends across institutions. Third, 2025 represents the most recent year for which disclosure information was available at the time of data collection, which includes the latest reporting period. As a result, the study develops three research hypotheses concerning the dynamics of ESG disclosure quality among three East Asian public pension funds.

P1: Do the ESG disclosure quality among the three major East Asian public pension funds exhibits systematic cross-national variation under PFDQI?

P2: Do the three pension funds display significant variation in their respective ESG disclosure practices?

P3: Is there evidence of a convergence trend in ESG disclosure quality among the pension funds of China, Japan, and South Korea?

4 Data analysis and results

4.1 PFDQI analyzation

This section provides a systematic assessment of the ESG disclosure quality of China's National Council for Social Security Fund (NCSSF), South Korea's National Pension Service (NPS), and Japan's Government Pension Investment Fund (GPIF) over the period 2020–2025. Table 4 reports the annual PFDQI scores for each institution, together with the corresponding cross-sectional means, standard deviations, and coefficients of variation (CV). The results indicate a clear and continuous improvement in ESG disclosure quality across all three pension funds during the observation period. Despite this common upward trajectory, substantial cross-national disparities persisted throughout the study period, revealing a pronounced hierarchical pattern of disclosure development. Among the three institutions, GPIF consistently achieved the highest level of ESG disclosure quality, with an average PFDQI score of 88.0 points. NPS occupied an intermediate position, recording a mean score of 63.9 points, while NCSSF exhibited the lowest overall disclosure quality, with an average score of 44.5 points.

Although notable differences remained, the rate of improvement varied considerably across institutions. NCSSF recorded the largest increase in PFDQI score during the study period, improving by 33.3 points between 2020 and 2025. This substantial increase suggests the presence of a catch-up dynamic, whereby pension funds with lower initial disclosure levels experienced more rapid improvement over time. By contrast, GPIF, which began the period with an already highly developed disclosure system, exhibited more moderate incremental gains, reflecting the diminishing marginal scope for improvement at advanced stages of disclosure development.

Table 4. Annual PFDQI Scores of Pension Funds across Three Countries (2020–2025)

Year	NCSSF (China)	NPS (South Korea)	GPIF (Japan)	Annual Mean	Std. Dev.	CV
2020	27.8	50.0	77.8	51.9	25.3	0.49
2021	33.3	55.6	83.3	57.4	25.0	0.44
2022	38.9	61.1	88.9	63.0	25.0	0.40
2023	50.0	66.7	88.9	68.5	19.8	0.29
2024	55.6	72.2	94.4	74.1	19.5	0.26
2025	61.1	77.8	94.4	77.8	16.8	0.22
Mean	44.5	63.9	88.0	65.4	—	—
Improvement	+33.3	+27.8	+16.6	—	—	—

Note: CV (Coefficient of Variation) = Standard Deviation / Cross-sectional Mean, reflecting the relative degree of dispersion among the three countries; improvement is calculated as the 2025 score minus the 2020 score.

Further analysis reveals a gradual narrowing of cross-national disparities over the period. The gap between GPIF and NCSSF remained relatively stable at approximately 50 points during 2020–2022 but subsequently declined to 33.3 points by 2025. A more pronounced reduction was observed in the GPIF–NPS differential, which decreased from 27.8 points in 2020 to 16.6 points in 2025. These findings suggest that although ESG disclosure quality among East Asian pension funds became progressively less polarized, substantial differences persisted throughout the observation period.

4.2 Dimensional Analysis of ESG Disclosure Quality

Table 5 reports the 2025-dimensional scores together with the corresponding score changes over the period 2020–2025. By disaggregating overall disclosure quality into its constituent dimensions, the analysis identifies the relative strengths and weaknesses of each institution and explores the specific dimensions that contribute to cross-national differences in ESG disclosure performance.

Table 5. Dimensional PFDQI Scores of Pension Funds across Three Countries (2025)

Dimension	NCSSF	NPS	GPIF	Score Differential	Full Score
Transparency	75.0	100.0	100.0	25.0	100
Completeness	62.5	75.0	100.0	37.5	100
Climate Disclosure	37.5	62.5	100.0	62.5	100
Responsible Investment (RI)	50.0	75.0	100.0	50.0	100
Credibility	50.0	50.0	75.0	25.0	100
Composite PFDQI	61.1	77.8	94.4	33.3	100

Note: Score differential refers to the gap between GPIF and NCSSF, capturing the extreme range of cross-country

disparity; the full score for each dimension is 100 points.

4.2.1 Transparency

The transparency dimension evaluates the extent to which pension funds publicly disclose ESG reports, responsible investment reports, ESG strategies, and measurable sustainability targets. As shown in Table 4-2, substantial differences were observed among the three institutions. GPIF achieved a full transparency score throughout the entire observation period, reflecting its long-standing commitment to comprehensive ESG reporting and its position as a global leader in responsible investment disclosure. NPS also demonstrated significant progress, attaining a perfect transparency score in 2022. This improvement coincided with the implementation of South Korea's revised Stewardship Code and the strengthening of ESG-related disclosure requirements for institutional investors. Among the three pension funds, NCSSF exhibited the most substantial improvement in transparency, with its score increasing from 25 points in 2020 to 75 points in 2025. This result reflects a notable expansion of ESG-related disclosure and increasing attention to responsible investment communication.

4.2.2 Completeness

The completeness dimension evaluates the extent to which pension funds systematically disclose information across the environmental (E), social (S), and governance (G) pillars, together with ESG performance indicators. GPIF consistently achieved a full score in this dimension, supported by comprehensive quantitative reporting on ESG performance, including portfolio carbon intensity and external ESG assessments. NPS also demonstrated relatively strong performance, particularly in the social dimension, reflecting the public welfare mandate of Korea's pension system. However, governance-related disclosures did not become fully institutionalized until 2023. In contrast, NCSSF exhibited comparatively lower completeness due to the limited quantitative depth of its ESG disclosures. Although environmental and social issues were increasingly discussed, reporting remained largely qualitative and lacked a standardized indicator framework aligned with international disclosure standards such as GRI and ISSB. As a result, the comparability and decision-usefulness of its disclosures remained relatively constrained (Berg et al., 2022).

4.2.3 Climate Disclosure

The climate disclosure dimension exhibits the greatest cross-national variation among the five PFDQI dimensions, with a score differential of 62.5 points in 2025. As shown in Table 6, GPIF consistently maintained a leading position by fully adopting the TCFD framework and conducting comprehensive climate scenario analyses covering multiple temperature pathways, including 1.5°C, 2°C, and 4°C scenarios. These disclosures were further supported by assessments of both physical and transition risks within its investment portfolio.

NPS demonstrated steady progress in climate disclosure following its endorsement of the TCFD framework in 2021. By 2024, the institution had incorporated multi-scenario climate analysis into its reporting practices. Nevertheless, its overall climate disclosure development remained several years behind that of GPIF.

In contrast, climate-related disclosure by NCSSF remained limited prior to 2023. Although climate information disclosure expanded following the advancement of China's green finance policies, quantitative climate-risk assessment and scenario analysis remained relatively underdeveloped as of 2025. These findings suggest that climate disclosure continues to represent the most significant source of variation in ESG disclosure quality among East Asian pension funds, reflecting substantial differences in institutional capacity and regulatory maturity. This result is consistent with Krueger et al. (2020), who find that climate-related disclosure among institutional investors in emerging markets generally lags behind that of developed-market counterparts.

Table 6. Indicators in the Climate Information Disclosure Dimension

Indicator	NCSSF	NPS	GPIF	Adoption Gap
TCFD Framework Adoption	2023	2021	2020	—
Climate Risk Analysis (Substantive)	2024	2022	2020	4-year gap
Carbon Reduction Targets (Quantitative)	2024	2022	2020	4-year gap
Climate Scenario Analysis (Multi-scenario)	Not completed	2024	2021	—
Dimension Score (2025)	37.5	62.5	100.0	62.5-point gap

Note: The adoption year for each indicator refers to the year in which substantive disclosure content first appeared in

that institution's publicly available annual report. "Not completed" indicates that systematic disclosure had not been achieved as of 2025.

4.2.4 Responsible Investment

The responsible investment dimension evaluates the disclosure of PRI reporting, stewardship activities, shareholder voting records, and corporate engagement practices. GPIF consistently achieved a full score in this dimension, supported by comprehensive reporting on voting outcomes, engagement activities, and responsible investment performance, reflecting a highly institutionalized responsible investment framework.

NPS also recorded substantial progress, with its score increasing from 50 points in 2020 to 75 points in 2025 following the adoption of its Responsible Investment Policy and the expansion of stewardship-related disclosures. However, disclosure of engagement activities at the individual case level remained relatively limited.

By contrast, NCSSF continued to lag behind in this dimension. While responsible investment principles received increasing attention, public disclosure of shareholder voting records and corporate engagement activities remained limited. This finding suggests that responsible investment disclosure among Chinese public pension funds remains at an early stage of development, particularly with respect to active ownership and stewardship practices (Clark & Hebb, 2004).

4.2.5 Credibility

The credibility dimension evaluates the adoption of third-party assurance and external verification mechanisms. Compared with the other PFDQI dimensions, all three pension funds exhibited relatively weak performance in this area. GPIF introduced limited external assurance in 2022 and achieved a score of 75 points in 2025, although reasonable assurance-level verification has yet to be implemented.

Both NPS and NCSSF recorded scores of 50 points in 2025, reflecting the limited use of independent assurance mechanisms. The consistently low scores across all three institutions suggest that external verification remains an underdeveloped component of ESG disclosure governance among East Asian public pension funds. This finding is consistent with Simnett et al. (2009), who report that ESG assurance adoption in the Asia-Pacific region generally lags behind that of Europe. Consequently, the limited development of assurance practices continues to constrain the overall credibility and reliability of ESG disclosures within the region.

4.3 Sigma-Convergence Analysis

To evaluate whether ESG disclosure quality among the three East Asian public pension funds exhibits a convergence tendency over time, this study employs a sigma-convergence (σ -convergence) analysis. Following the convergence literature, cross-sectional standard deviation (σ) and the coefficient of variation (CV) are used to measure changes in the dispersion of PFDQI scores across institutions (Barro & Sala-i-Martin, 1992; Islam, 2003). A decline in either indicator over time suggests that differences in ESG disclosure quality are narrowing, thereby providing evidence of convergence.

Table 7 reports the annual values of σ and CV for the period 2020–2025 and serves as the basis for assessing the convergence dynamics of ESG disclosure quality among the NCSSF, NPS, and GPIF.

Indicator	2020	2021	2022	2023	2024	2025
Standard Deviation (σ)	25.3	25.0	25.0	19.8	19.5	16.8
Coefficient of Variation (CV)	0.49	0.44	0.40	0.29	0.26	0.22
GPIF–NCSSF Gap	50.0	50.0	50.0	38.9	38.8	33.3
GPIF–NPS Gap	27.8	27.7	27.8	22.2	22.2	16.6

Note: Standard deviation (σ) denotes the cross-sectional standard deviation of annual PFDQI scores across the three funds; Coefficient of Variation (CV) = σ / cross-sectional mean; Gap refers to the absolute score difference. A sustained decline in σ is the criterion for confirming the presence of σ -convergence.

As shown in Table 7, both the cross-sectional standard deviation (σ) and the coefficient of variation (CV) declined steadily over the study period. Specifically, σ decreased from 25.3 in 2020 to 16.8 in 2025, while CV fell from 0.49 to 0.22, indicating

a clear sigma-convergence trend in ESG disclosure quality among the three East Asian pension funds.

However, this convergence should be interpreted with caution. The sharpest decline in dispersion occurred between 2022 and 2023, coinciding with the accelerated improvement of NCSSF, suggesting that convergence was driven primarily by a catch-up effect among lower-performing institutions. Despite the narrowing gap, substantial differences remained: the score differential between GPIF and NCSSF still reached 33.3 points in 2025, indicating that convergence remained partial rather than complete.

Moreover, convergence was uneven across disclosure dimensions. Differences in transparency and completeness narrowed more rapidly than those in climate disclosure and credibility, implying that advanced disclosure capabilities—particularly climate-risk assessment, scenario analysis, and external assurance—face higher institutional barriers to development. Overall, the findings suggest a pattern of gradual but uneven convergence, in which persistent institutional differences continue to shape ESG disclosure outcomes. This result is consistent with Liang and Renneboog (2017), who argue that the diffusion of ESG practices across countries remains strongly conditioned by national institutional contexts.

Overall, three major findings are concluded from the analysis.

First, the ESG disclosure quality of all three pension funds improved substantially during 2020–2025, although a clear three-tier structure persisted. GPIF maintained a leading position, NPS demonstrated steady progress, and NCSSF exhibited the most pronounced catch-up effect. These findings provide strong support for P1, confirming the existence of systematic cross-national differences in ESG disclosure quality among East Asian public pension funds.

Second, climate disclosure and credibility remained the weakest dimensions across the three institutions. While climate disclosure was constrained by the technical and organizational demands of climate-risk assessment and scenario analysis, credibility was limited by the relatively underdeveloped adoption of external assurance mechanisms within the East Asian pension fund sector. These results provide evidence for P2, indicating substantial variation across the core dimensions of ESG disclosure quality.

Third, the sigma-convergence results indicate a gradual narrowing of cross-national differences in ESG disclosure quality. However, this convergence was driven primarily by improvements among lower-performing institutions and remains incomplete. Significant disclosure gaps therefore continue to persist despite the overall convergence trend. These findings support P3 by demonstrating the existence of a partial convergence process among the pension funds of China, Japan, and South Korea.

5 Conclusion

This study developed a Pension Fund Disclosure Quality Index (PFDQI) to evaluate the ESG disclosure quality of three major East Asian public pension funds—China's NCSSF, South Korea's NPS, and Japan's GPIF—over the period 2020–2025. By combining content analysis with sigma-convergence analysis, the study examined both cross-national variation and the dynamic evolution of ESG disclosure quality. From a theoretical perspective, the findings suggest that ESG disclosure quality functions not only as an indicator of responsible investment maturity but also as a reflection of broader institutional governance capacity. The study further demonstrates that convergence in ESG disclosure does not necessarily imply institutional homogenization, as national governance contexts continue to exert a significant influence on disclosure practices.

Furthermore, the findings of this study generate several important policy implications for regulators, pension fund managers, and sustainable finance policymakers in East Asia. First, regulators should promote the development of ESG disclosure frameworks specifically tailored to pension funds. Although international standards such as GRI, TCFD, ISSB, and PRI provide important guidance, greater standardization is needed to enhance the comparability and consistency of pension fund disclosures across countries.

Second, climate-risk disclosure should be strengthened as a core component of pension fund governance. Given that climate disclosure remains the dimension with the greatest cross-national variation, pension funds should improve their capabilities in climate-risk assessment, target setting, and scenario analysis to enhance long-term risk management and

investment decision-making. Third, greater transparency should be encouraged in responsible investment practices. Enhanced disclosure of stewardship activities, shareholder voting records, and corporate engagement outcomes would strengthen accountability and improve stakeholder confidence in pension fund governance. Finally, policymakers should support the development of independent assurance mechanisms for ESG information. Expanding the use of third-party verification would improve the credibility and reliability of disclosures and contribute to higher-quality ESG governance across the pension fund sector.

At the end, several limitations should be acknowledged. First, the study focuses on three representative East Asian public pension funds, which limits the generalizability of the findings. Future research may extend the analysis to pension funds in Europe, North America, and other regions. Second, although content analysis provides a systematic approach to evaluating disclosure quality, some degree of subjective judgment remains unavoidable. Future studies may incorporate empirical study to improve measurement objectivity.

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